Campus Lost & Found

A working local prototype

THE TEAM

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Notice workflow

Data analysis

Simulated smart lockers

The student workflow

A fictional found notebook, followed through the working application

01

Publish

Add a location and an optional photo

02

Find

Search saved notices and open the item

03

Contact

Open an email draft to the notice author

04

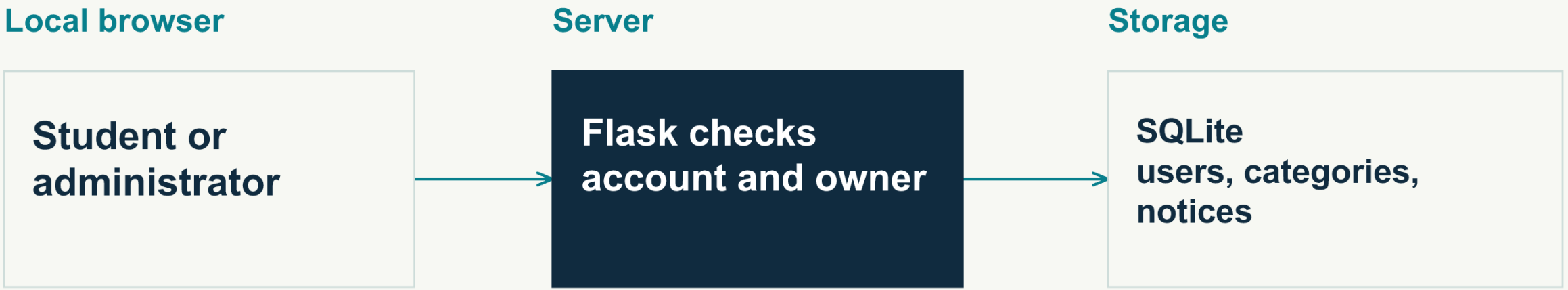
Return

The owner marks the notice returned

70-second live demo

SQLite keeps each saved change.
The contact action opens a draft email.

Access and data boundaries



Protected changes

Salted scrypt passwords
CSRF tokens and role checks

Private photo uploads

JPG or PNG, up to 5 MB
Server validates and re-encodes

University identity

Domain restriction only
SSO and mailbox verification pending

Current deployment: local loopback HTTP

Review by five subagents

AI subagents reviewed separate parts of the local prototype

SUBAGENT	REVIEW SCOPE	BASELINE
AI-01	Student workflow and media	15
AI-02	Registration and private uploads	11
AI-03	Database and analytics	21
AI-04	Research recorder and simulated locker	14
AI-05	Backup and recovery	4

72

final automated checks passed

65 baseline checks
+ 7 regression checks

2 issues fixed

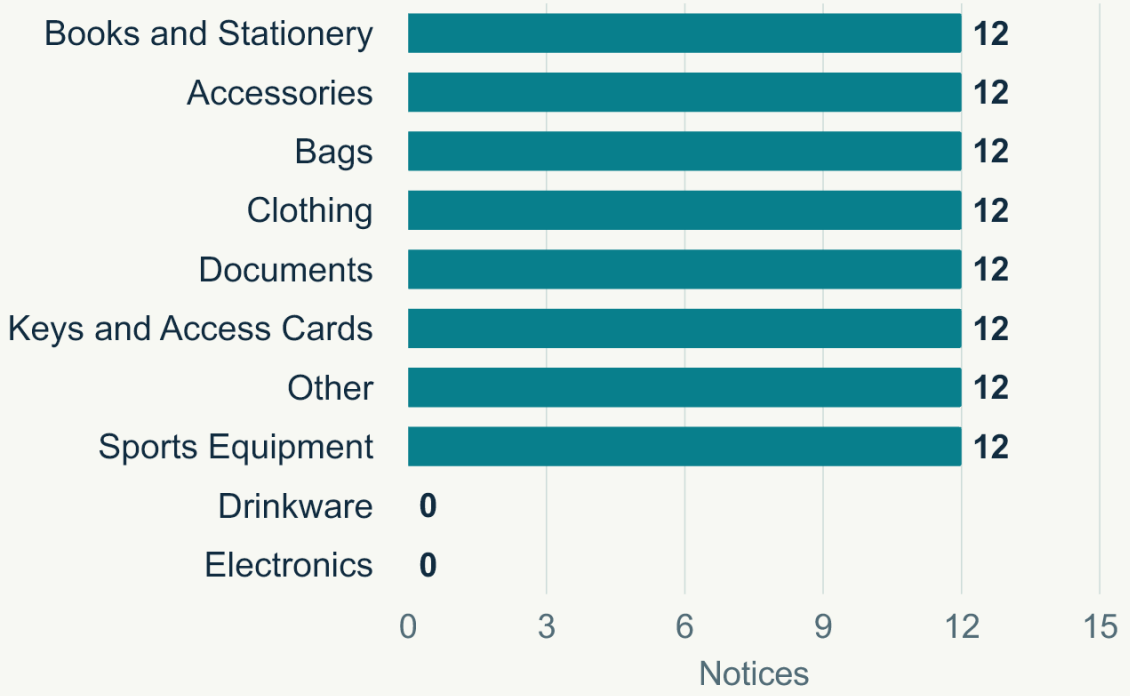
External human task study: awaiting five participants.

Category and location patterns

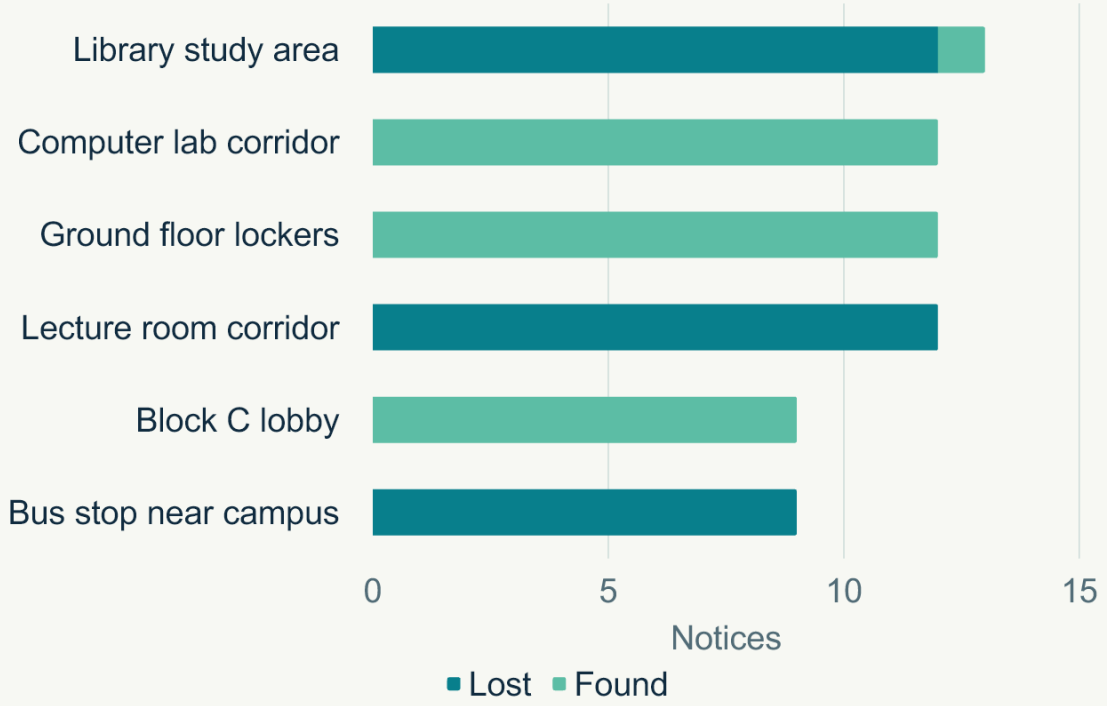
122 synthetic notices

96 open / 26 returned

Open notices by category



Top six recorded locations

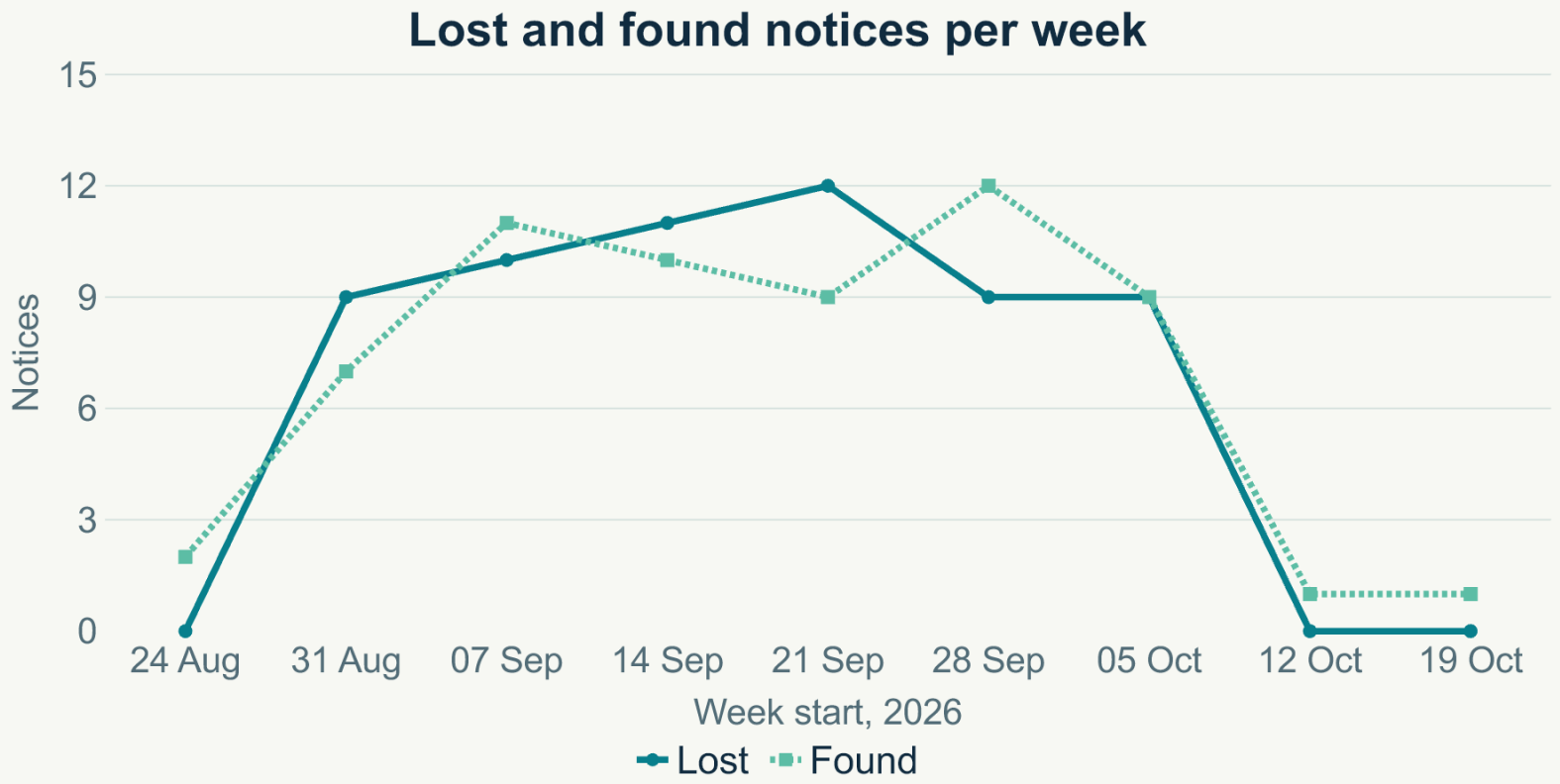


Seed bias: all 12 Electronics and 13 Drinkware notices are marked returned.

Synthetic snapshot: 30 Oct 2026, 10:34 UTC. Location ties use alphabetic order.

Weekly patterns and a design decision

Monday–Sunday weeks, grouped by item event date



DESIGN DECISION

Standardise location choices

Free-form labels can split equivalent places.

20-second live SQL

Run the saved category query.

Synthetic data. Event date does not measure return time. Live demo edits may change the figures.

Campus IoT in practice

University of Birmingham

23,000

Enlighted sensors in lighting fixtures

Occupancy and movement
Temperature and energy

Responsive lighting and room-use planning

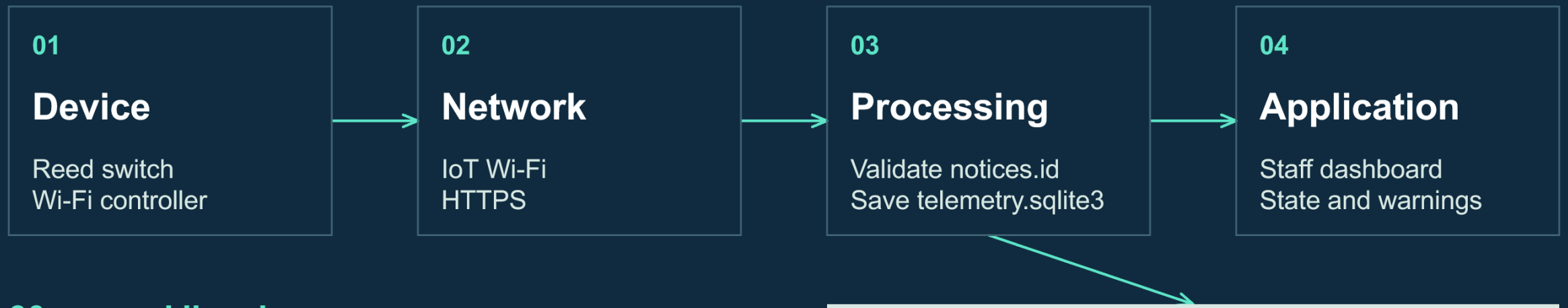


Linked occupancy histories can reveal routines. Campus hosting is unspecified.

A smart-locker extension

Four layers for a proposed physical pilot

SIMULATED



30-second live demo

Open for 10 seconds, refresh for a warning.
Close the simulated door to clear it.

No hardware. No automatic item return.

30 seconds

Proposed physical open rule

Illustrative estimate

14,600 records/day

10 lockers × 1,460 events × 200 bytes = 2.92 MB/day payload + overhead

Before submission and Week 10

Week 9

Complete the evidence

Five external participants
Timed publish and find tasks
Browser, Fast 4G and offline checks

Week 10

Fix, retest and rehearse

Address observed usability problems
Repeat affected checks
Rehearse the 6:30 team defense

Mengdigerey Zhanzhol

Application and access

Galitskiy Daniil

Tests and data analysis

Bisengaliev Daulet

IoT and demonstration

University SSO and remote TLS remain deployment work.